



Mutual Fund Analytics

Capstone Project

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INTERNSHIP: BLUESTOCK FINTECH

TOOLS USED: PYTHON, SQL, SQLITE, POWER BI

YEAR: 2026

Problem Statement & Objective

Problem Statement

- Mutual fund data is scattered across multiple sources.
- Fund comparison is difficult due to complex metrics.
- Benchmark tracking is not available.
- Investor behaviour analysis is limited.
- Manual reporting is slow and time-consuming.

Objectives

- Integrate all datasets into one database.
- Analyse fund performance and risk.
- Compare funds with benchmark indices.
- Study investor investment trends.
- Build interactive Power BI dashboards.

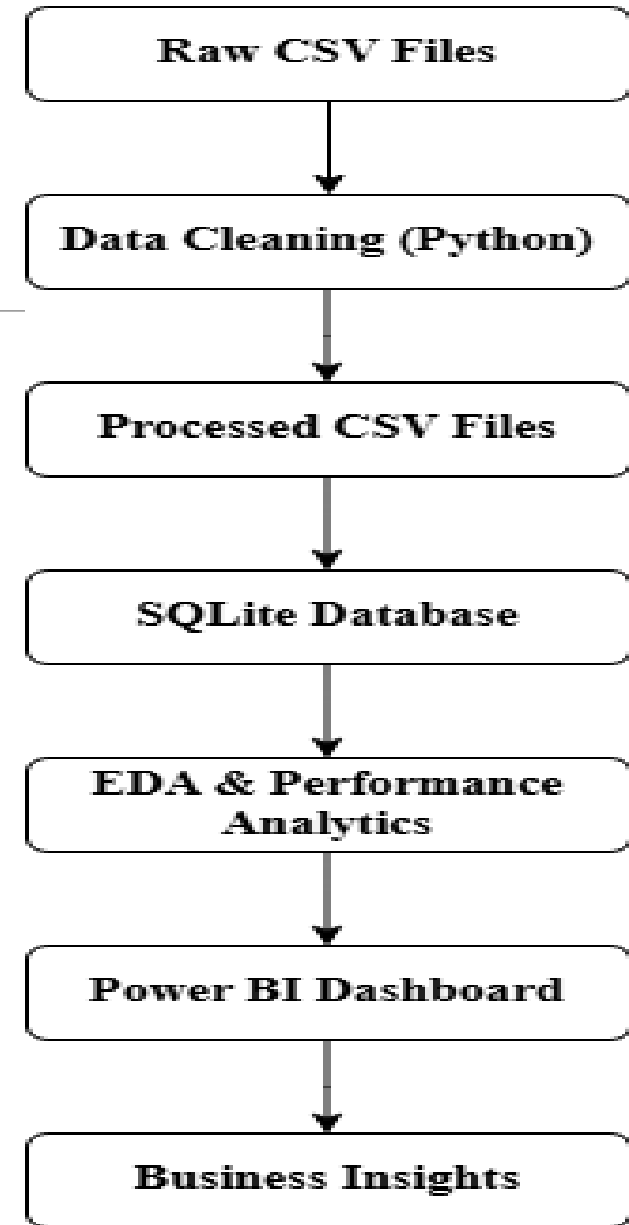
Data Sources

Dataset	Description
Fund Master	Fund details and category information
NAV History	Daily Net Asset Value (NAV) records
AUM	Assets Under Management of fund houses
SIP	Monthly Systematic Investment Plan inflows
Category	Category-wise investment inflows
Folio	Industry folio (investor account) count
Transactions	Investor purchase and redemption details
Performance	Fund return and performance metrics
Portfolio	Portfolio holdings of mutual funds
Benchmark	Nifty 50 and Nifty 100 market index data

Project Architecture

Tools Used

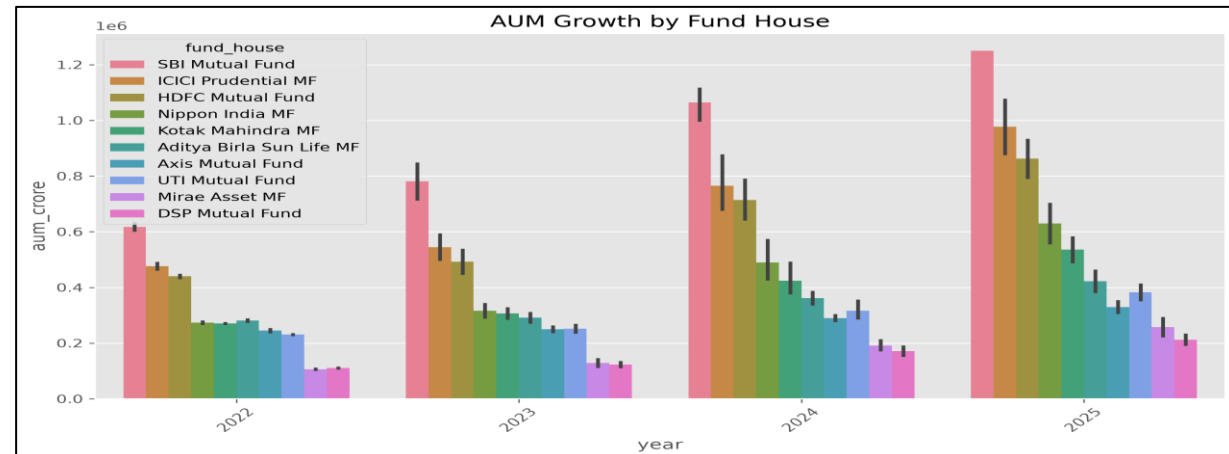
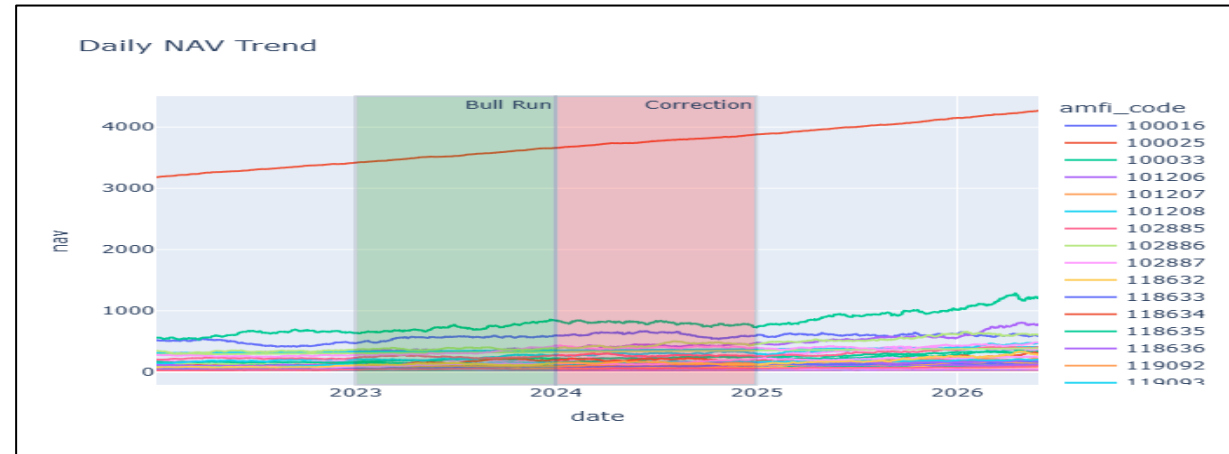
- Python
- Pandas
- SQLAlchemy
- SQLite
- Power BI
- Git & GitHub



EDA Highlights - Industry

Key Findings

- NAV increased steadily.
- AUM showed continuous growth.
- SIP investments increased yearly.
- Investor participation improved.
- Industry showed positive growth.



Key Findings

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- Monthly SIP Inflow Trend**
- The chart displays the monthly SIP inflow in Crores (Cr) from April 2022 to March 2026. The y-axis represents the inflow in Crores, ranging from 10k to 30k. The x-axis represents the month, spanning from 2022 to 2026. The inflow starts at approximately 11,500 Cr in April 2022, remains relatively flat until mid-2023, then shows a steady upward trend, reaching a peak of ₹31,002 Cr in March 2026.
- | Month | SIP Inflow (Cr) |
|---------|-----------------|
| 2022-04 | 11,500 |
| 2022-05 | 11,800 |
| 2022-06 | 12,000 |
| 2022-07 | 12,200 |
| 2022-08 | 12,500 |
| 2022-09 | 12,800 |
| 2022-10 | 13,000 |
| 2022-11 | 13,200 |
| 2022-12 | 13,500 |
| 2023-01 | 13,800 |
| 2023-02 | 14,000 |
| 2023-03 | 14,200 |
| 2023-04 | 14,500 |
| 2023-05 | 14,800 |
| 2023-06 | 15,000 |
| 2023-07 | 15,200 |
| 2023-08 | 15,500 |
| 2023-09 | 15,800 |
| 2023-10 | 16,000 |
| 2023-11 | 16,200 |
| 2023-12 | 16,500 |
| 2024-01 | 16,800 |
| 2024-02 | 17,000 |
| 2024-03 | 17,200 |
| 2024-04 | 17,500 |
| 2024-05 | 17,800 |
| 2024-06 | 18,000 |
| 2024-07 | 18,200 |
| 2024-08 | 18,500 |
| 2024-09 | 18,800 |
| 2024-10 | 19,000 |
| 2024-11 | 19,200 |
| 2024-12 | 19,500 |
| 2025-01 | 19,800 |
| 2025-02 | 20,000 |
| 2025-03 | 20,200 |
| 2025-04 | 20,500 |
| 2025-05 | 20,800 |
| 2025-06 | 21,000 |
| 2025-07 | 21,200 |
| 2025-08 | 21,500 |
| 2025-09 | 21,800 |
| 2025-10 | 22,000 |
| 2025-11 | 22,200 |
| 2025-12 | 22,500 |
| 2026-01 | 22,800 |
| 2026-02 | 23,000 |
| 2026-03 | 23,200 |
| 2026-04 | 23,500 |
| 2026-05 | 23,800 |
| 2026-06 | 24,000 |
| 2026-07 | 24,200 |
| 2026-08 | 24,500 |
| 2026-09 | 24,800 |
| 2026-10 | 25,000 |
| 2026-11 | 25,200 |
| 2026-12 | 25,500 |
| 2027-01 | 25,800 |
| 2027-02 | 26,000 |
| 2027-03 | 26,200 |
| 2027-04 | 26,500 |
| 2027-05 | 26,800 |
| 2027-06 | 27,000 |
| 2027-07 | 27,200 |
| 2027-08 | 27,500 |
| 2027-09 | 27,800 |
| 2027-10 | 28,000 |
| 2027-11 | 28,200 |
| 2027-12 | 28,500 |
| 2028-01 | 28,800 |
| 2028-02 | 29,000 |
| 2028-03 | 29,200 |
| 2028-04 | 29,500 |
| 2028-05 | 29,800 |
| 2028-06 | 30,000 |
| 2028-07 | 30,200 |
| 2028-08 | 30,500 |
| 2028-09 | 30,800 |
| 2028-10 | 31,002 |

Performance Metrics

Financial Metrics Calculated

- **Daily Returns** – Measured daily fund performance.
- **CAGR (1, 3 & 5 Years)** – Calculated long-term annual growth.
- **Sharpe Ratio** – Measured return compared to total risk.
- **Sortino Ratio** – Evaluated return considering downside risk only.
- **Alpha** – Measured fund performance against the benchmark.
- **Beta** – Measured fund sensitivity to market movements.
- **Maximum Drawdown** – Calculated the largest loss from peak value.
- **Fund Scorecard** – Compared mutual funds based on overall performance.

Advanced Analytics & Risk Analysis

Advanced Analytics

Risk Metrics Calculated

- **Value at Risk (VaR)** – Estimated the maximum expected loss at a given confidence level.
- **Conditional Value at Risk (CVaR)** – Measured the average loss beyond the VaR limit.
- **Rolling 90-Day Sharpe Ratio** – Evaluated changes in risk-adjusted returns over time.
- **Investor Cohort & SIP Analysis** – Studied investor behaviour and SIP continuity.
- **Fund Recommendation System** – Ranked mutual funds based on return, risk, and performance.

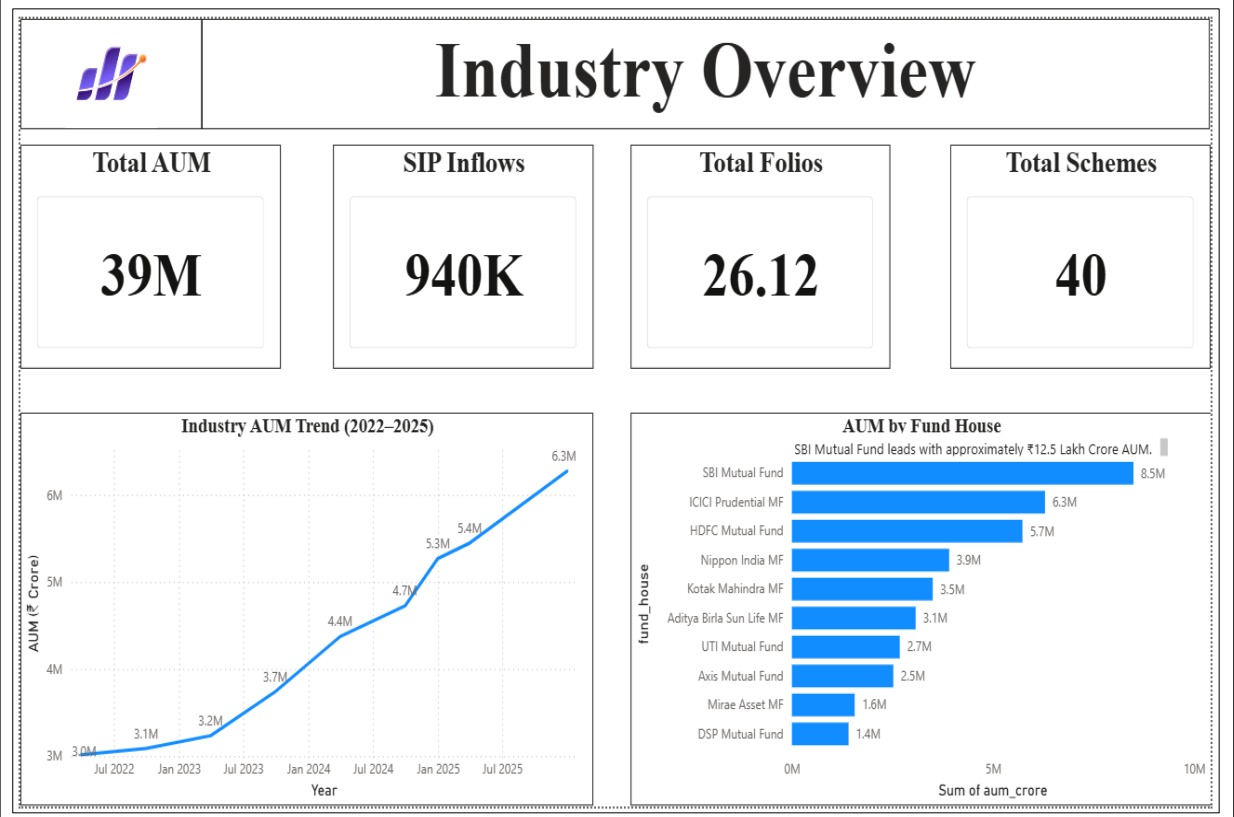
Benefits

- Identified high-risk and low-risk mutual funds.
- Measured downside risk more accurately.
- Analysed investor investment patterns.
- Compared funds using advanced financial metrics.
- Helped investors choose suitable mutual fund schemes.

Dashboard 1 & 2

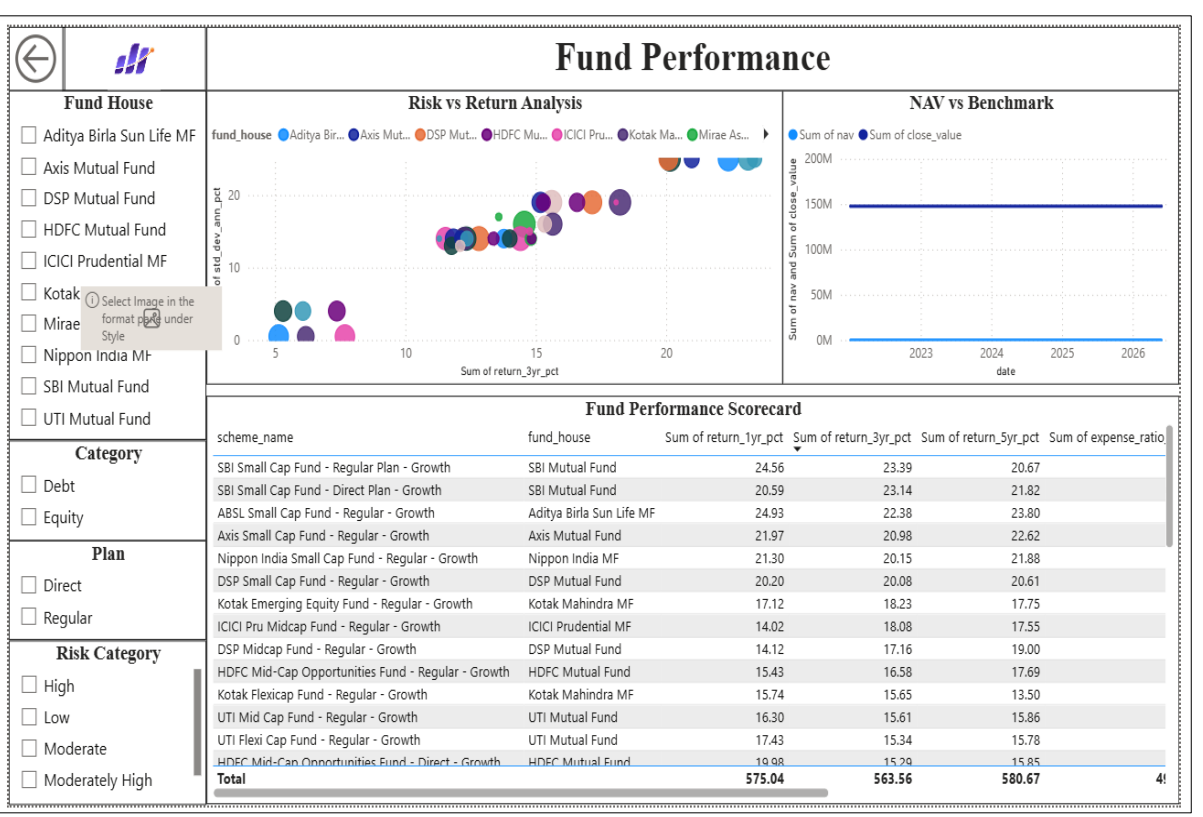
Industry Overview

1.Total AUM 2.Total SIP 3.Total Folios 4.Industry Growth



Fund Performance

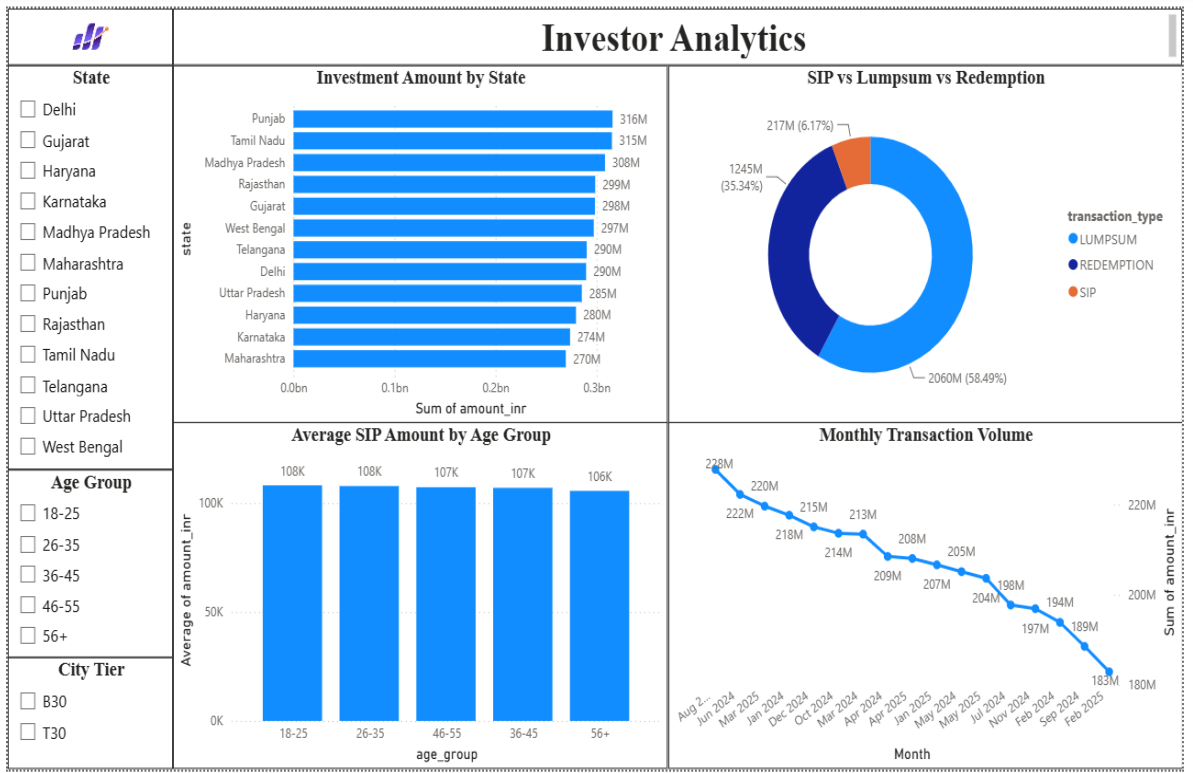
1.Risk vs Return 2.NAV Trend 3.Benchmark Comparison
4.Fund Filters



Dashboard 3 & 4

Investor Analytics

- 1.State-wise Investments
- 2.Age Group
- 3.AnalysisTransactions
- 4.Monthly Trends



Key Findings

- Mutual fund **AUM and SIP inflows** showed consistent growth, indicating increasing investor participation.
- **Equity funds** delivered better long-term returns compared to debt funds.
- **Large-cap funds** demonstrated more stable performance with lower risk.
- Performance metrics such as **CAGR, Sharpe Ratio, Alpha, and Beta** helped identify the best-performing funds.
- Interactive **Power BI dashboards** simplified fund comparison and supported better investment decisions.

Thank You